

Summative Assessment 1

DSC1105 Exploratory Data Analysis

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GitHub Link:

<https://github.com/ChewyGnome/DSC1105>

Dataset

The dataset that will be used in this report is the csv file “EDA Ecommerce Assessment”. This contains the information of the behavior of how consumers act when making a transaction in an e-commerce platform. Variables such as the number of items purchased, amount of transaction, browsing time, date of transaction, consumer satisfaction scores are all given in the dataset. This dataset also contains demographic variables such as gender and age, which is taken into consideration of analyzing consumer demands.

The objective of this analysis is to find the common patterns that the consumers practice when making a transaction. This analysis will employ exploration methods such as univariate and bivariate to identify the variables that enable a consumer in making a purchase.

```
ecommerce <- read.csv("EDA_Ecommerce_Assessment.csv")
```

Data Inspection

Structure of the Dataset

```
class(ecommerce)
```

```
## [1] "data.frame"
```

First 6 observations of the dataset

```
head(ecommerce)
```

```
## Customer_ID Gender Age Browsing_Time Purchase_Amount Number_of_Items
## 1 1 Male 65 46.55 231.81 6
## 2 2 Female 19 98.80 472.78 8
## 3 3 Male 23 79.48 338.44 1
## 4 4 Male 45 95.75 37.13 7
## 5 5 Male 46 33.36 235.53 3
## 6 6 Female 43 83.39 123.92 9
## Discount_Applied Total_Transactions Category Satisfaction_Score
## 1 17 16 Clothing 2
## 2 15 43 Books 4
## 3 28 31 Electronics 1
## 4 43 27 Home & Kitchen 5
## 5 10 33 Books 3
## 6 5 29 Clothing 2
```

Summary statistics of the dataset

```
summary(ecommerce)
```

```
## Customer_ID Gender Age Browsing_Time
## Min. : 1.0 Length:3000 Min. :18.00 Min. : 1.00
## 1st Qu.: 750.8 Class :character 1st Qu.:31.00 1st Qu.: 29.98
## Median :1500.5 Mode :character Median :44.00 Median : 59.16
## Mean :1500.5 Mean :43.61 Mean : 59.87
## 3rd Qu.:2250.2 3rd Qu.:57.00 3rd Qu.: 89.33
## Max. :3000.0 Max. :69.00 Max. :119.95
## Purchase_Amount Number_of_Items Discount_Applied Total_Transactions
## Min. : 5.03 Min. :1.00 Min. : 0.00 Min. : 1.00
## 1st Qu.:128.69 1st Qu.:3.00 1st Qu.:12.00 1st Qu.:12.00
## Median :245.09 Median :5.00 Median :24.00 Median :24.00
## Mean :247.96 Mean :4.99 Mean :24.34 Mean :24.68
## 3rd Qu.:367.20 3rd Qu.:7.00 3rd Qu.:37.00 3rd Qu.:37.00
## Max. :499.61 Max. :9.00 Max. :49.00 Max. :49.00
## Category Satisfaction_Score
## Length:3000 Min. :1.000
## Class :character 1st Qu.:2.000
## Mode :character Median :3.000
## Mean :3.066
## 3rd Qu.:4.000
## Max. :5.000
```

Dataset structure

```
str(ecommerce)
```

```
## 'data.frame': 3000 obs. of 10 variables:
## $ Customer_ID : int 1 2 3 4 5 6 7 8 9 10 ...
## $ Gender : chr "Male" "Female" "Male" "Male" ...
## $ Age : int 65 19 23 45 46 43 42 29 22 51 ...
```

```
## $ Browsing_Time      : num  46.5 98.8 79.5 95.8 33.4 ...
## $ Purchase_Amount    : num  231.8 472.8 338.4 37.1 235.5 ...
## $ Number_of_Items    : int   6 8 1 7 3 9 6 5 8 8 ...
## $ Discount_Applied   : int   17 15 28 43 10 5 2 13 1 31 ...
## $ Total_Transactions : int   16 43 31 27 33 29 8 33 41 19 ...
## $ Category           : chr   "Clothing" "Books" "Electronics" "Home & Kitchen" ...
## $ Satisfaction_Score : int    2 4 1 5 3 2 1 3 4 5 ...
```

Note: This dataset uses a mixture of both categorical and numerical variables. Variables such as Browsing_Time, Purchase_Amount, Number_of_Items, and Satisfaction_Score are numerical, while variables such as Gender and Category are categorical. These variables will be used in this report as the primary basis in observing customer behavior.

Exploratory Data Analysis

Variable Identification in the Dataset

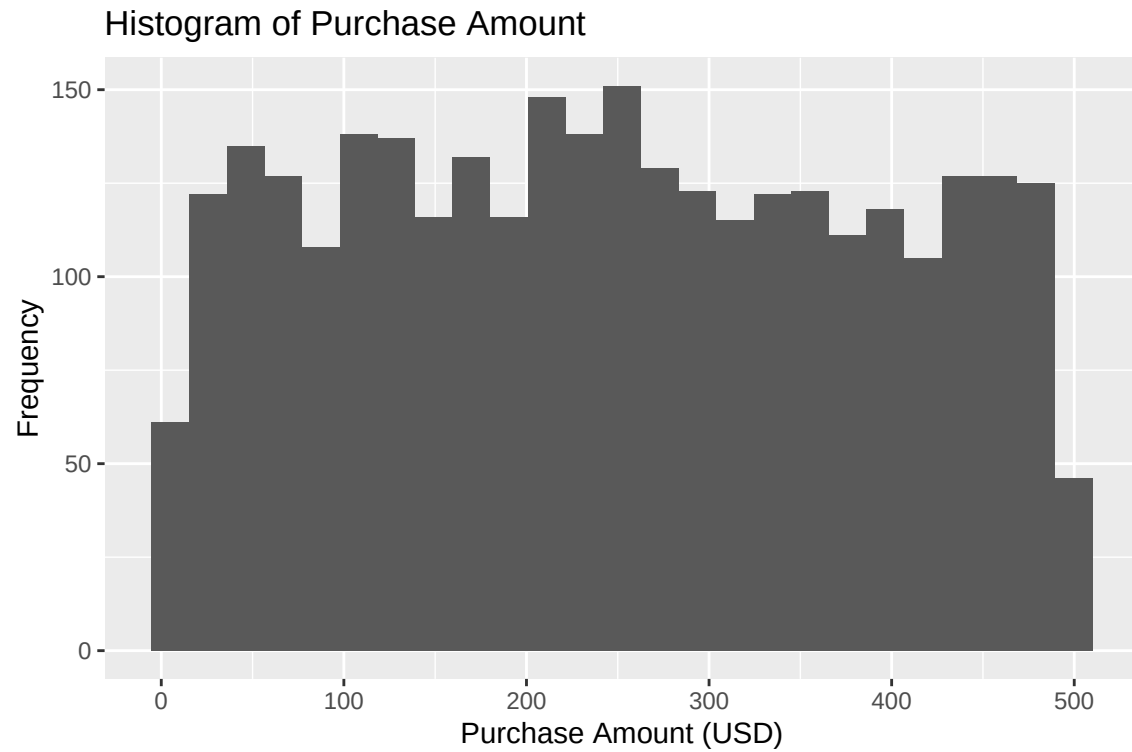
Response variable: Purchase_Amount (Amount spent in a transaction)

Primary explanatory variable: Browsing_Time (Amount of time spent before making a purchase)

Additional explanatory variables includes: - Number_of_Items - Satisfaction_Score - Gender - Category

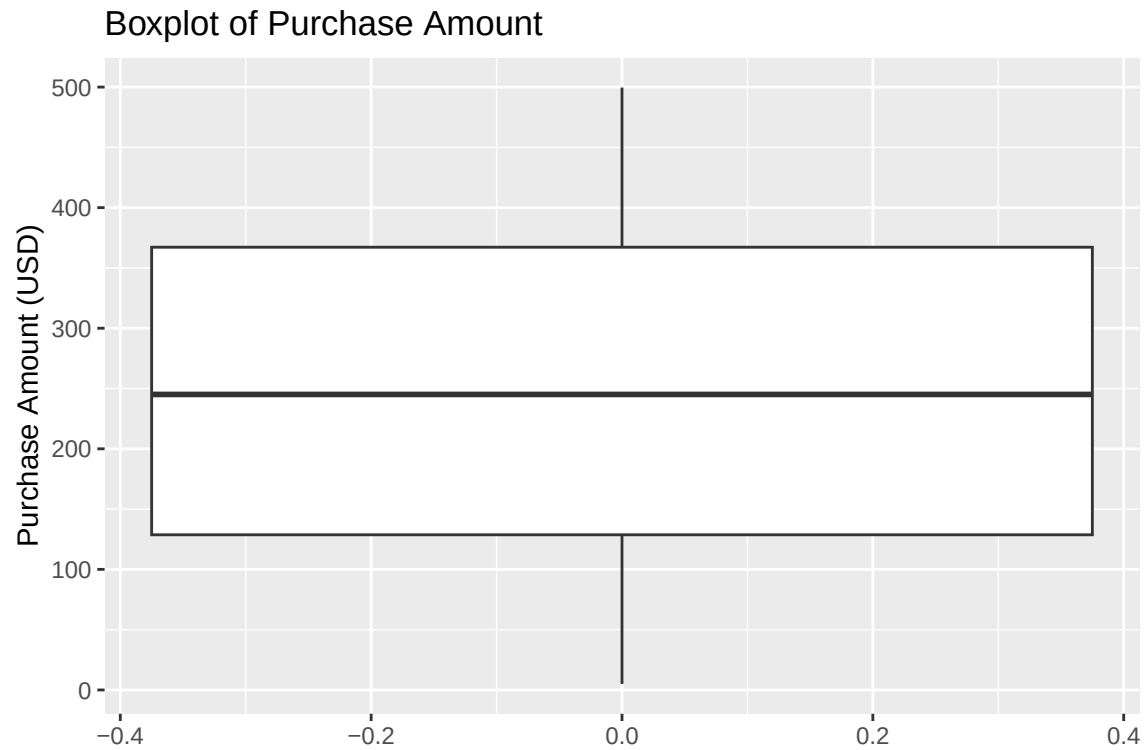
Histogram of Purchase Amount

```
ggplot(ecommerce, aes(x = Purchase_Amount)) +
  geom_histogram(bins = 25) +
  labs(
    title = "Histogram of Purchase Amount",
    x = "Purchase Amount (USD)",
    y = "Frequency"
  )
```



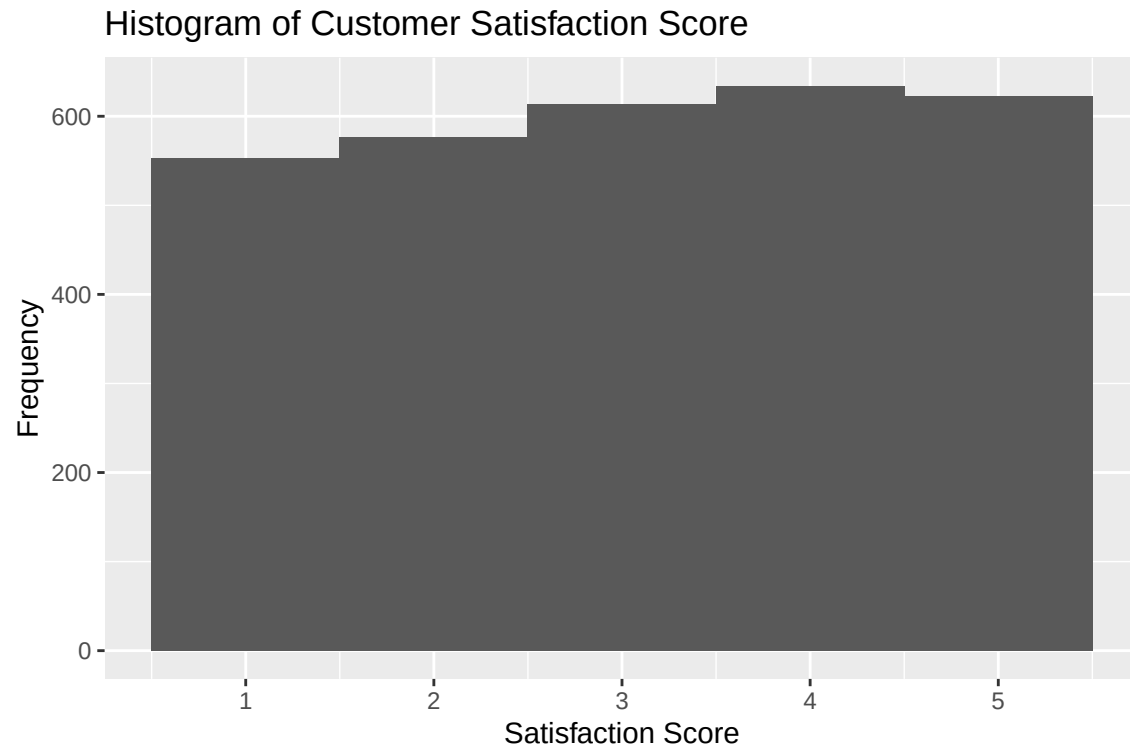
Boxplot of Purchase Amount

```
ggplot(ecommerce, aes(y = Purchase_Amount)) +  
  geom_boxplot() +  
  labs(  
    title = "Boxplot of Purchase Amount",  
    y = "Purchase Amount (USD)"  
  )
```



Histogram of Number of Items Purchased

```
ggplot(ecommerce, aes(x = Satisfaction_Score)) +  
  geom_histogram(bins = 5) +  
  labs(  
    title = "Histogram of Customer Satisfaction Score",  
    x = "Satisfaction Score",  
    y = "Frequency"  
  )
```



Measures of Central Tendency and Spread

```
mean(ecommerce$Purchase_Amount)
```

```
## [1] 247.9625
```

```
median(ecommerce$Purchase_Amount)
```

```
## [1] 245.09
```

```
mode_value <- names(sort(table(ecommerce$Purchase_Amount), decreasing = TRUE))[1]  
mode_value
```

```
## [1] "29.33"
```

```
var(ecommerce$Purchase_Amount)
```

```
## [1] 19845.99
```

```
sd(ecommerce$Purchase_Amount)
```

```
## [1] 140.8758
```

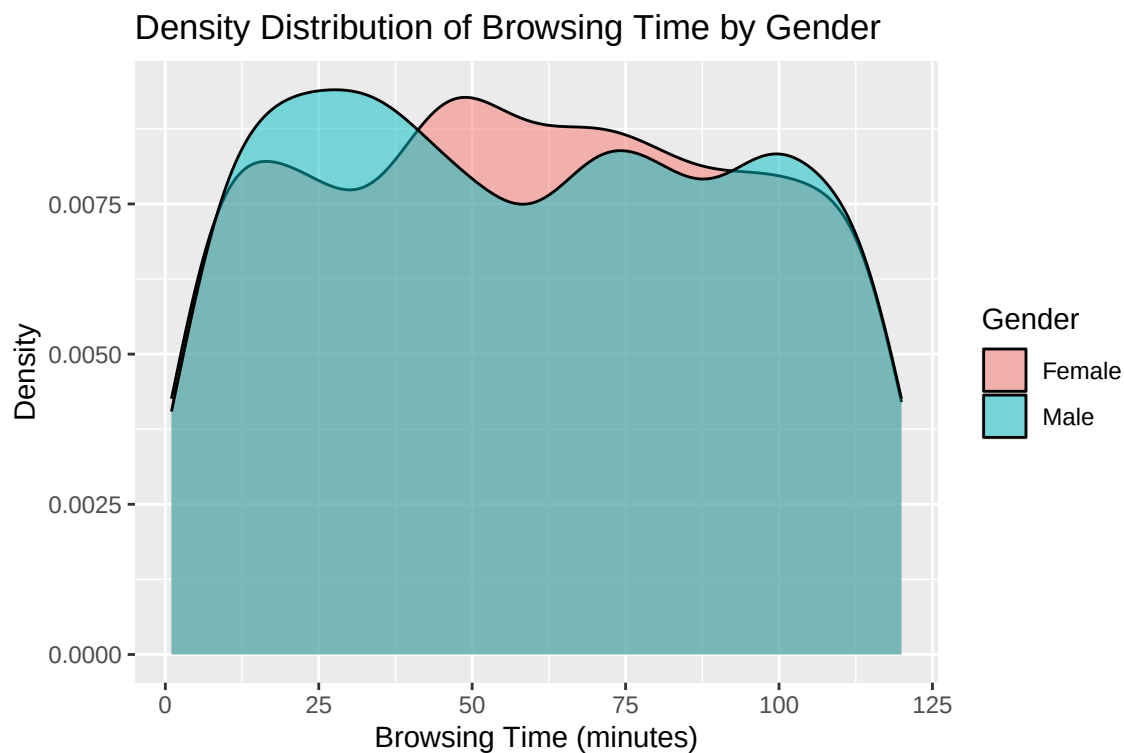
```
IQR(ecommerce$Purchase_Amount)
```

```
## [1] 238.505
```

Density Comparison by Gender

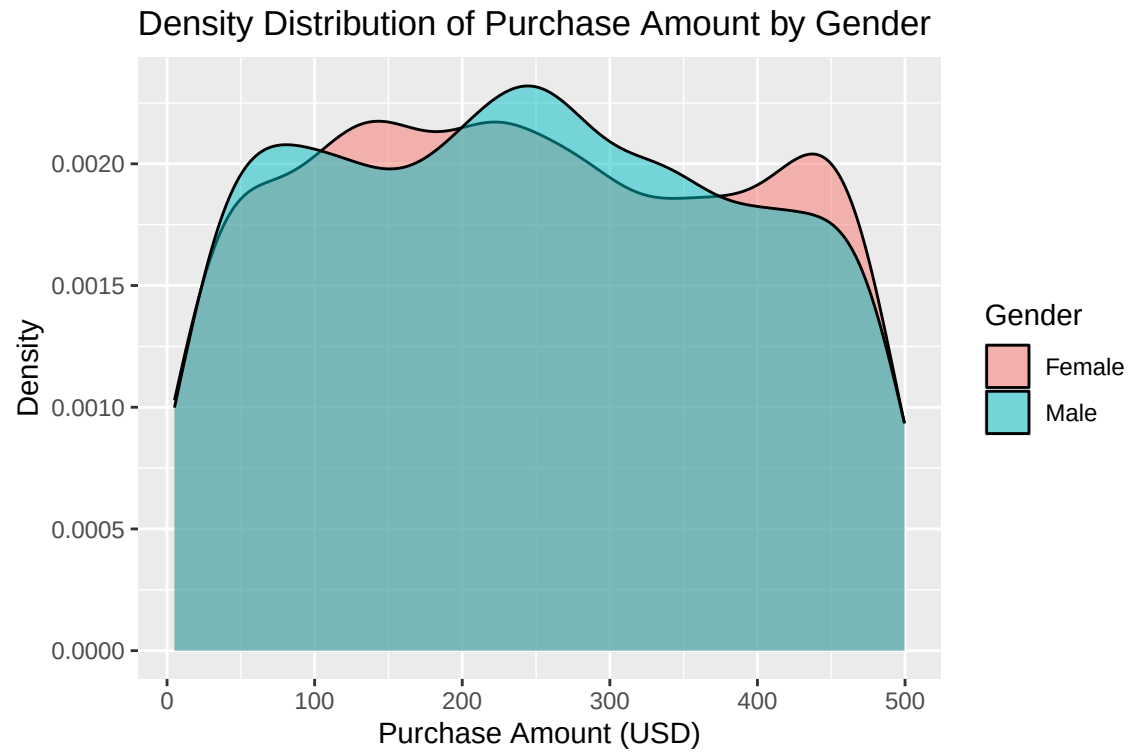
Browsing Time by Gender

```
ggplot(ecommerce, aes(x = Browsing_Time, fill = Gender)) +  
  geom_density(alpha = 0.5) +  
  labs(  
    title = "Density Distribution of Browsing Time by Gender",  
    x = "Browsing Time (minutes)", y = "Density"  
  )
```



Purchase Amount by Gender

```
ggplot(ecommerce, aes(x = Purchase_Amount, fill = Gender)) +  
  geom_density(alpha = 0.5) +  
  labs(  
    title = "Density Distribution of Purchase Amount by Gender",  
    x = "Purchase Amount (USD)",  
    y = "Density"  
  )
```

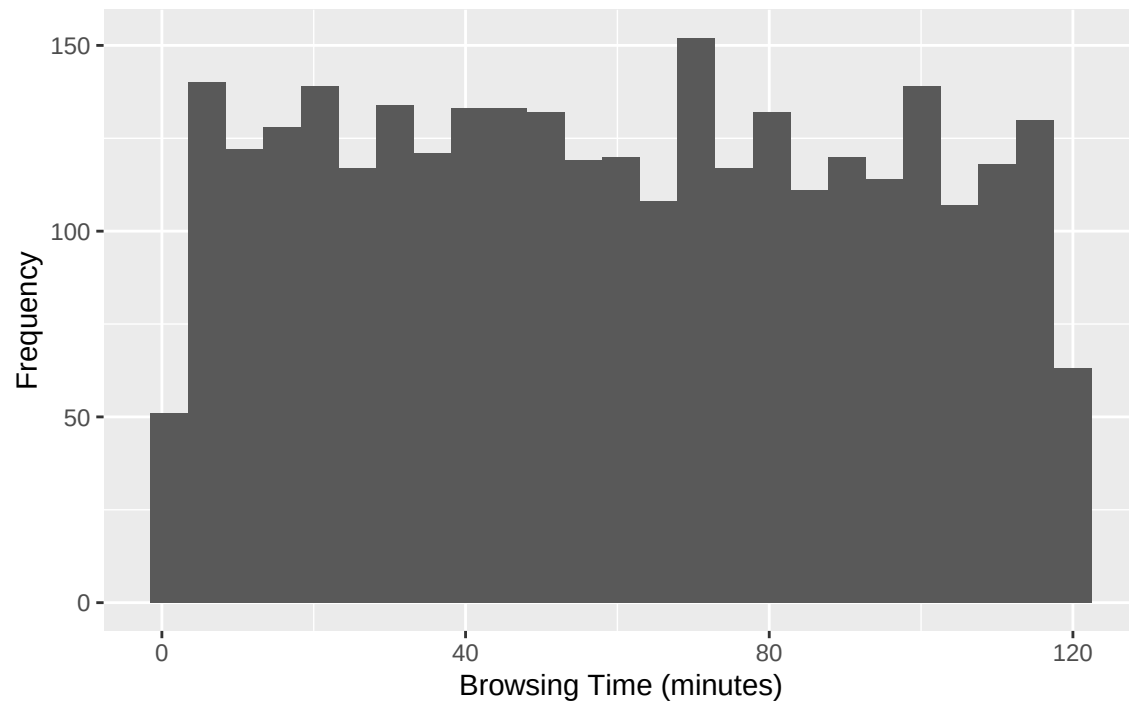


Transformation of Browsing Time

Original Distribution

```
ggplot(ecommerce, aes(x = Browsing_Time)) +  
  geom_histogram(bins = 25) +  
  labs(  
    title = "Original Distribution of Browsing Time",  
    x = "Browsing Time (minutes)",  
    y = "Frequency"  
  )
```


Original Distribution of Browsing Time

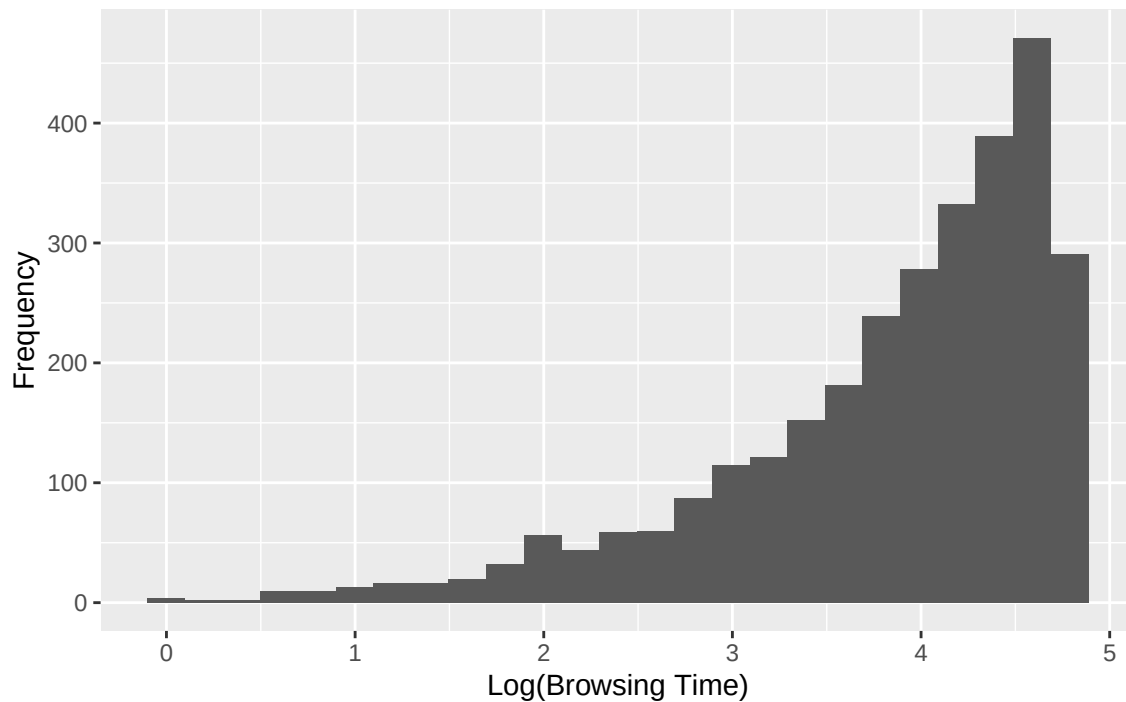


Log Transformation of Browsing Time

```
ecommerce$log_browsing <- log(ecommerce$Browsing_Time)

ggplot(ecommerce, aes(x = log_browsing)) +
  geom_histogram(bins = 25) +
  labs(
    title = "Log Transformed Distribution of Browsing Time",
    x = "Log(Browsing Time)",
    y = "Frequency"
  )
```

Log Transformed Distribution of Browsing Time



Simple Linear Regression Model

Model Fitting

```
purchase_lm <- lm(Purchase_Amount ~ Browsing_Time, data = ecommerce)
summary(purchase_lm)
```

```
##
## Call:
## lm(formula = Purchase_Amount ~ Browsing_Time, data = ecommerce)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -244.867 -120.473  -2.946  118.246  254.069
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  252.65596    5.17524  48.820  <2e-16 ***
## Browsing_Time -0.07839    0.07501  -1.045    0.296
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 140.9 on 2998 degrees of freedom
## Multiple R-squared:  0.0003642, Adjusted R-squared:  3.075e-05
## F-statistic: 1.092 on 1 and 2998 DF, p-value: 0.2961
```

Model Interpretation

The fitted linear regression model is expressed as

$$\text{Purchase_Amount} = \beta_0 + \beta_1(\text{Browsing_Time})$$

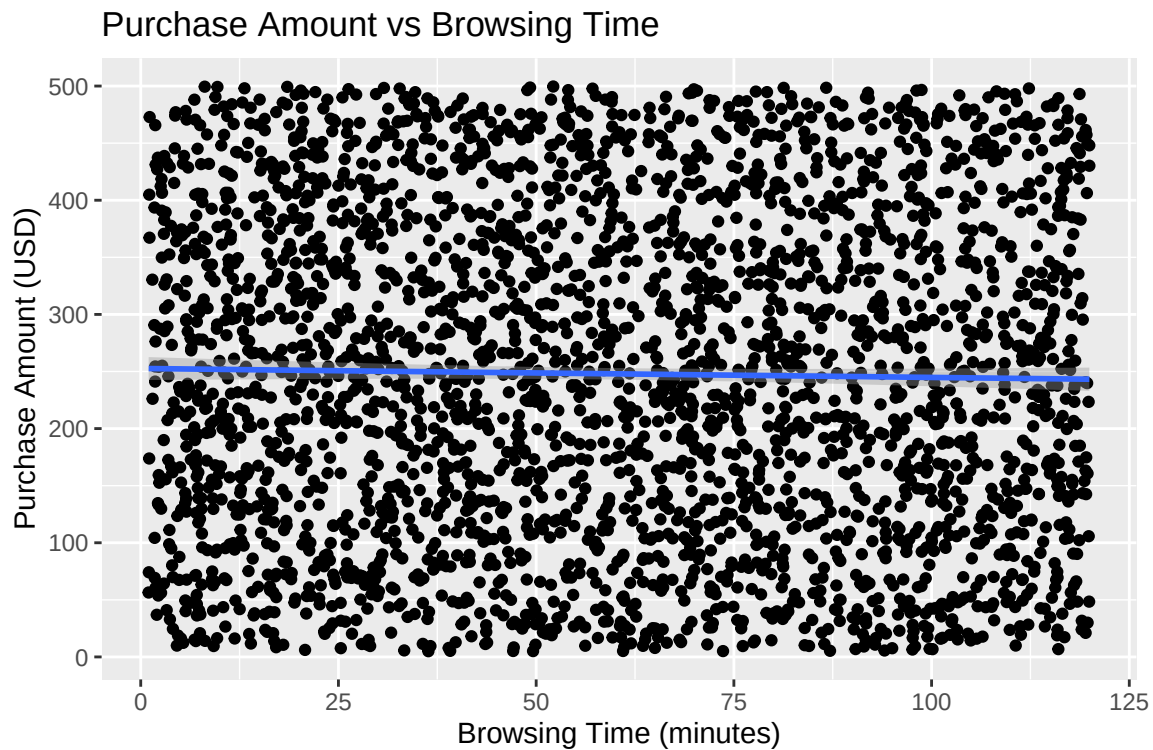
Slope: The estimated output for the slope shows the predicted change in purchase amount for every additional minute spent browsing the website before making a transaction.

Intercept: The intercept represents the predicted purchase amount when browsing time is set to zero minutes.

Scatter Plot of Purchase Amount vs Browsing Time

```
ggplot(ecommerce, aes(x = Browsing_Time, y = Purchase_Amount)) +  
  geom_point() +  
  geom_smooth(method = "lm") +  
  labs(  
    title = "Purchase Amount vs Browsing Time",  
    x = "Browsing Time (minutes)",  
    y = "Purchase Amount (USD)"  
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

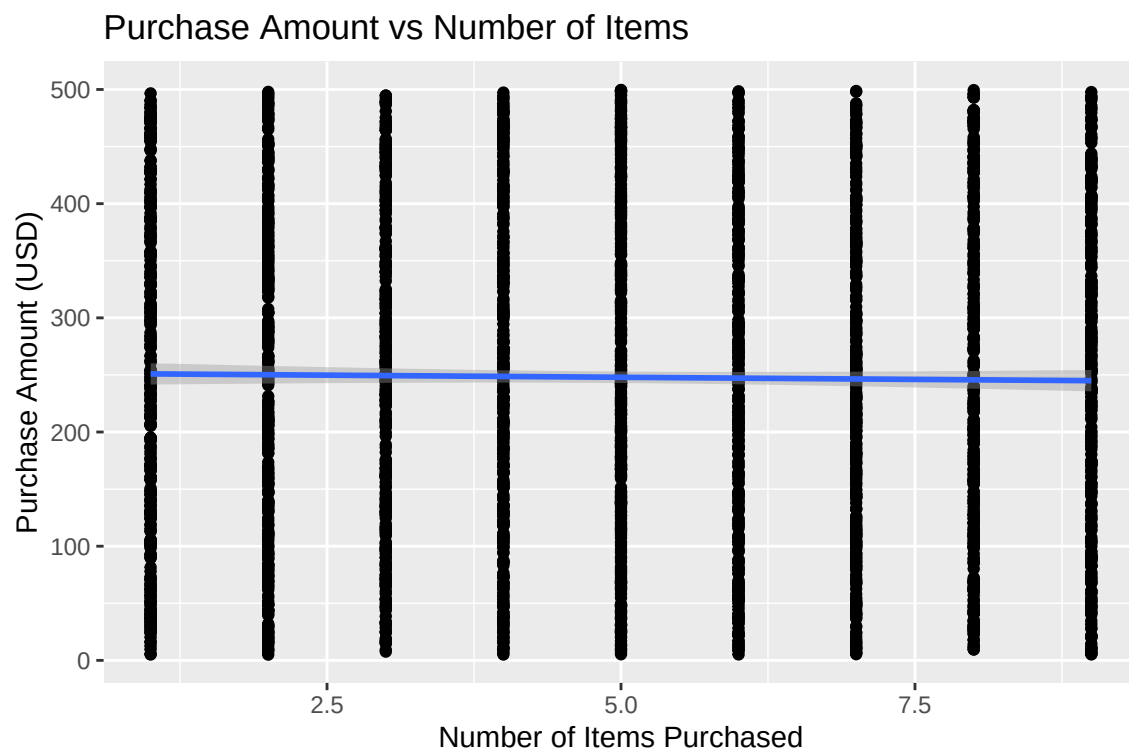


Bivariate Analysis

Purchase Amount vs Number of Items

```
ggplot(ecommerce, aes(x = Number_of_Items, y = Purchase_Amount)) +  
  geom_point() +  
  geom_smooth(method = "lm") +  
  labs(  
    title = "Purchase Amount vs Number of Items",  
    x = "Number of Items Purchased",  
    y = "Purchase Amount (USD)"  
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Polynomial Regression Model

```
poly_model <- lm(Purchase_Amount ~ poly(Browsing_Time, 2), data = ecommerce)  
summary(poly_model)
```

```
##  
## Call:  
## lm(formula = Purchase_Amount ~ poly(Browsing_Time, 2), data = ecommerce)  
##
```

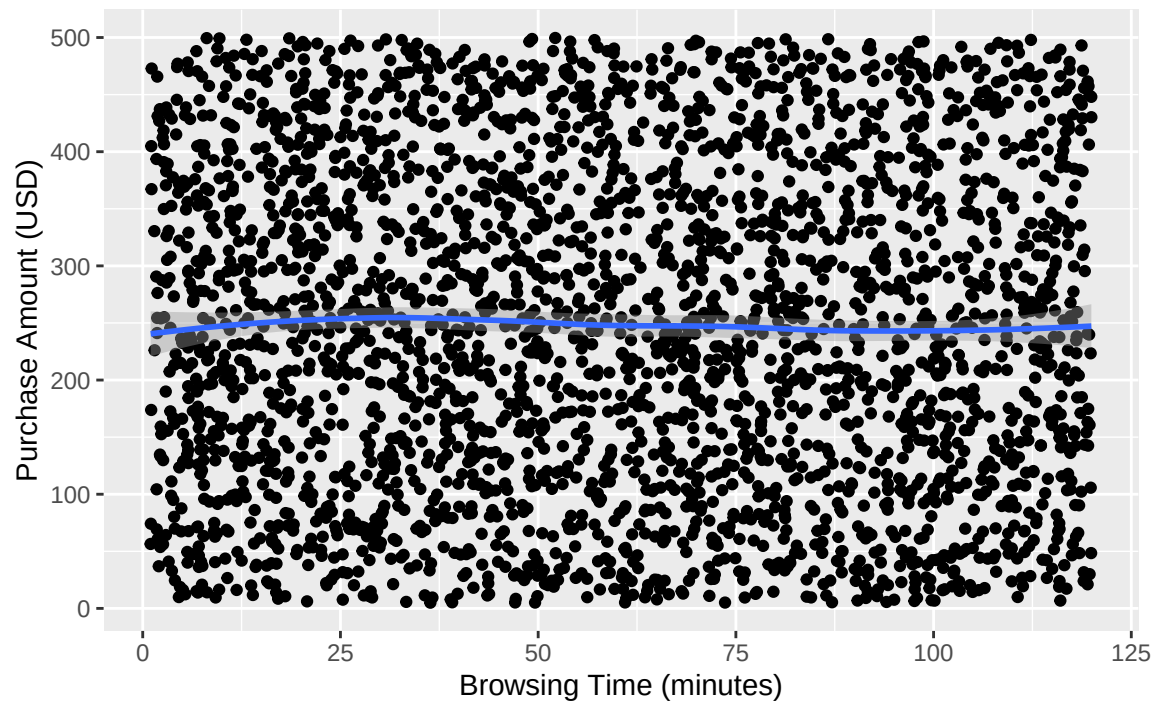
```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -245.47 -120.41   -3.49  118.25  255.85
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      247.963      2.572  96.397  <2e-16 ***
## poly(Browsing_Time, 2)1 -147.227    140.892  -1.045    0.296
## poly(Browsing_Time, 2)2  -68.129    140.892  -0.484    0.629
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 140.9 on 2997 degrees of freedom
## Multiple R-squared:  0.0004422, Adjusted R-squared:  -0.0002249
## F-statistic: 0.6629 on 2 and 2997 DF, p-value: 0.5154
```

LOESS Smoothing

```
ggplot(ecommerce, aes(x = Browsing_Time, y = Purchase_Amount)) +
  geom_point() +
  geom_smooth(method = "loess") +
  labs(
    title = "LOESS Smoothing of Purchase Amount vs Browsing Time",
    x = "Browsing Time (minutes)",
    y = "Purchase Amount (USD)"
  )
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

LOESS Smoothing of Purchase Amount vs Browsing Time



Robust Regression

```
robust_model <- rlm(Purchase_Amount ~ Browsing_Time, data = ecommerce)
summary(robust_model)
```

```
##
## Call: rlm(formula = Purchase_Amount ~ Browsing_Time, data = ecommerce)
## Residuals:
```

##	Min	1Q	Median	3Q	Max
##	-244.818	-120.331	-2.848	118.291	254.289

```
##
## Coefficients:
```

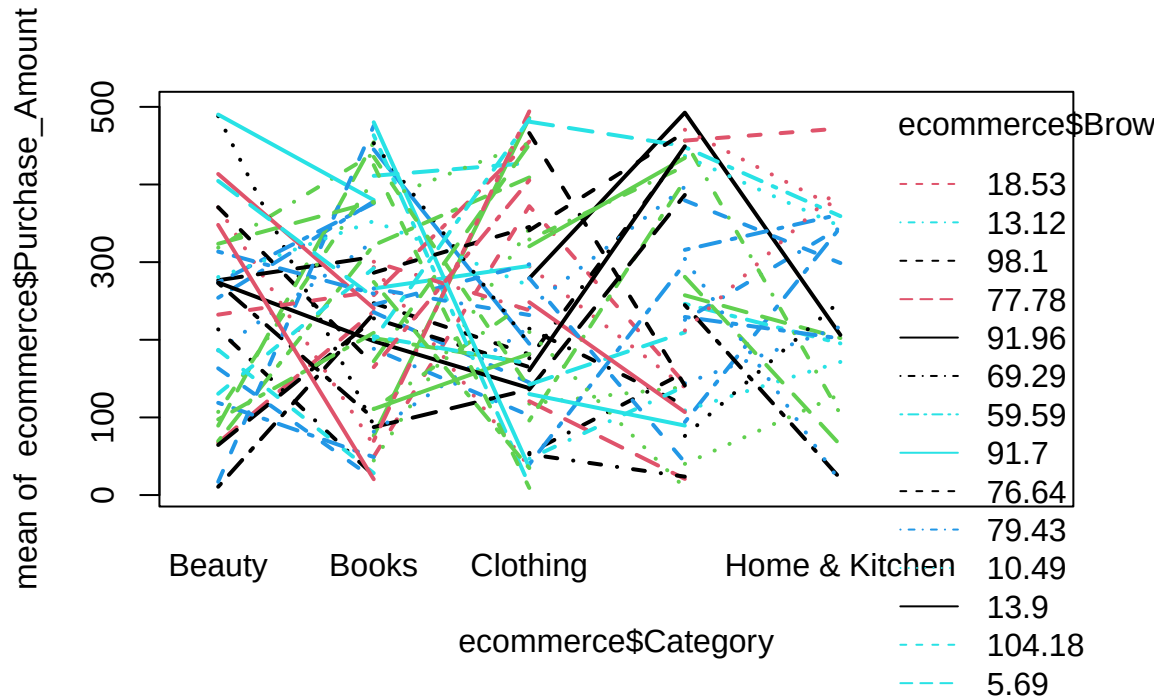
##		Value	Std. Error	t value
##	(Intercept)	252.6462	5.3363	47.3448
##	Browsing_Time	-0.0803	0.0773	-1.0378

```
##
## Residual standard error: 176.9 on 2998 degrees of freedom
```

Trivariate / Hypervariate Analysis

Interaction Plot

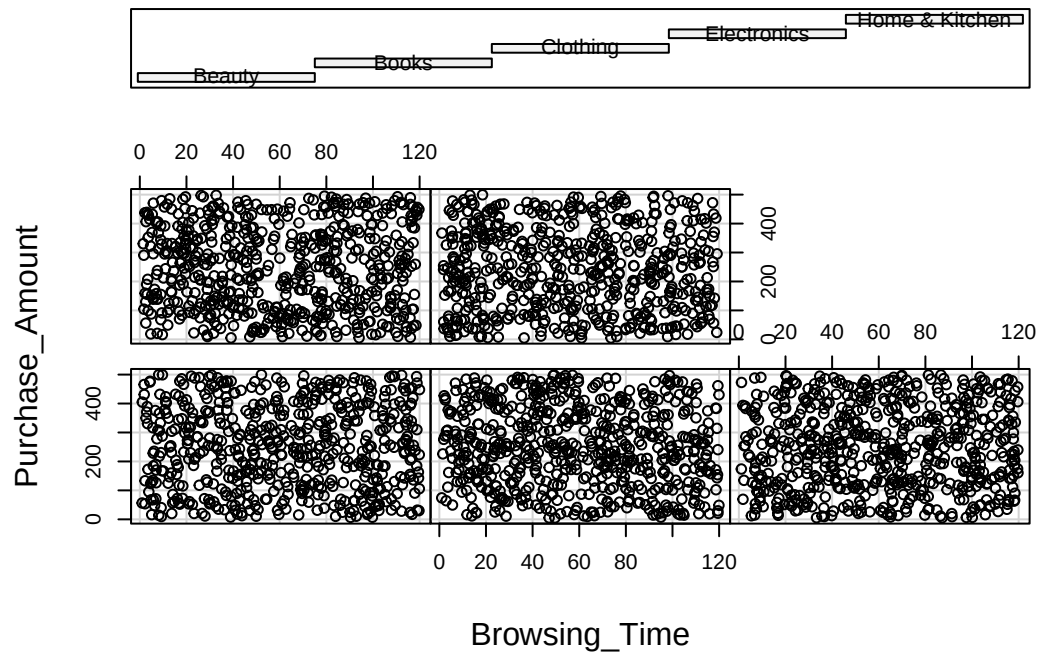
```
interaction.plot(
  ecommerce$Category,
  ecommerce$Browsing_Time,
  ecommerce$Purchase_Amount,
  col = 1:5,
  lwd = 2
)
```



Coplots

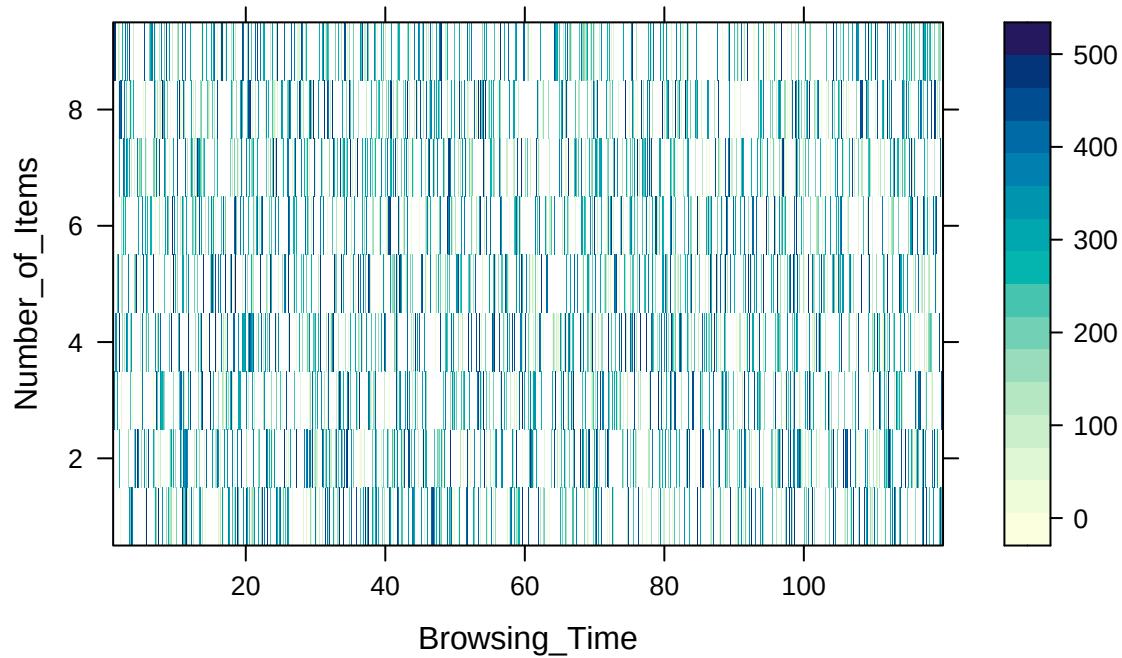
```
coplot(Purchase_Amount ~ Browsing_Time | Category, data = ecommerce)
```

Given : Category



Level Plot

```
levelplot(Purchase_Amount ~ Browsing_Time * Number_of_Items, data = ecommerce)
```

Multiple Regression Model

Model Fitting

```
multi_model <- lm(
  Purchase_Amount ~ Browsing_Time + Number_of_Items + Satisfaction_Score,
  data = ecommerce
)

summary(multi_model)
```

```
##
## Call:
## lm(formula = Purchase_Amount ~ Browsing_Time + Number_of_Items +
##     Satisfaction_Score, data = ecommerce)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -250.668 -120.856  -2.846  118.899  255.664
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    261.34993     9.24929   28.256  <2e-16 ***
## Browsing_Time    -0.07954     0.07504   -1.060    0.289
## Number_of_Items  -0.78321     1.00497   -0.779    0.436
## Satisfaction_Score -1.53871     1.83444   -0.839    0.402
```

```
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 140.9 on 2996 degrees of freedom  
## Multiple R-squared:  0.0007932, Adjusted R-squared:  -0.0002073  
## F-statistic: 0.7928 on 3 and 2996 DF,  p-value: 0.4978
```

Model Interpretation

In this multiple regression model variables such as browsing time, number of items purchased, and customer satisfaction are examined to make an accurate assumption in what factors influence purchase amounts.

Reflection

This report provided insightful information in regards to consumer behavior when making a transaction. Through the usage of plots, summaries and regression modeling several patterns were observed from consumer behavior in relation to their purchasing habits. From the results there is a direct correlation between the variable of “browsing time” and “purchase amount” this suggests that consumers get more incentive in making a purchase in relation to their time investment.

The usage of visualization techniques in modelling also helped reveal the distribution of variables and their correlation to one another. Additionally, applying a logarithmic transformation to browsing time allowed for a clearer understanding of its distribution and improved interpretability in the analysis. Another important factor is the regression analysis, fitting simple and multiple regression models making it possible to quantify how certain variables influence purchase behavior.

Concluding this, this activity reinforced the importance of interpretation in complex and multifaceted data. This shows how exploratory data analysis is a crucial first step in analyzing different datasets.